

Hei Shing Helson Go

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SUMMARY

State estimation and localization engineer with seven years building and validating real-time probabilistic robotics systems for autonomous vehicles. Specialist in multi-sensor fusion, Kalman filtering, GPS/GNSS-denied localization, and optimization-based estimation, in performance-critical modern **C++** and **Rust**. Proven track record taking estimation algorithms from mathematical formulation through high-fidelity simulation to deployment on physical vehicles, including real-time embedded targets.

LOCALIZATION & STATE ESTIMATION EXPERIENCE

University of Toronto Institute for Aerospace Studies

Sep 2020 – Oct 2025 | Toronto, ON

Ph.D. Researcher — Flight Systems & Control Lab

Doctoral research: cooperative, precision localization for GNSS-denied autonomous vehicles

- **Designed** real-time, optimization-based state estimation and planning algorithms delivering sub-0.5 m positioning accuracy without GPS for teams of collaborating and mutually sensing autonomous vehicles.
- **Established** a new mathematical link between system observability and achievable positioning precision, yielding an active sensing principle that shapes vehicle motion to maximize the information from sensors, cutting positioning error by **over 75%** versus baseline in Monte Carlo studies.
- **Validated** the localization pipeline end-to-end in simulation and on physical quadrotor platforms with inter-drone UWB trackers, reducing real-world positioning uncertainty by **54–61%** and holding it under 2 m where a conventional filter diverged past 4 m; published across three peer-reviewed venues.

Applanix (Trimble) collaboration: LiDAR-inertial odometry for GNSS-denied localization

- **Re-engineered** a modern C++20 LiDAR-inertial odometry library based on AriseLab *Super-Odometry* to complement Applanix's APX-15 GNSS system for GNSS-denied multirotor localization.
- **Implemented** a modular pipeline containing edge/planar feature extraction, IMU preintegration, scan-to-map registration (ICP), an iteratively-reweighted Gauss–Newton optimization that replaced *Ceres Solver*, and an octree local map that replaced *iKd-Tree*.
- **Flight-tested** the system in closed loop: streamed a **200 Hz** IMU-propagated odometry feed into *PX4*'s EKF2 and flew a drone in position-controlled mode on the LiDAR odometry as its sole position source; covered **248 m** autonomously at up to **6 m/s** without triggering EKF2 warnings.

Early-Stage Autonomous Drone Racing Venture

Oct 2025 – Present | Toronto, ON

Co-founder & Chief Engineer

- **Re-implemented** the racing team's autopilot as a real-time embedded flight controller in Rust while exploring commercialization of the platform, containing an Error-State Kalman Filter (ESKF) for multi-sensor fusion of IMU, barometer, magnetometer, and RTK GNSS for localization.
- **Defined** safety requirements for the flight controller, created a fault-tolerant sensor supervisor that detects and handles sensor faults, and validated it with equivalence tests and benchmarks.
- **Built** the firmware as a bare-metal `no_std` system on an ARM Cortex-M7 microcontroller; transitioned from indoor motion-capture to outdoor GNSS localization while maintaining **sub-0.3 m** positioning accuracy, and demonstrated a **108 km/h** top speed (Flight footage available on request).

University of Toronto Autonomous Drone Racing Team

Sep 2024 – 2025 | Toronto, ON

Engineering Lead

Competition autopilot: optimization- and learning-based high-speed autonomous flight

- **Led** a team of 7 building an optimization- and machine-learning-based autopilot for a high-speed autonomous drone, placing **6th worldwide** at the 2025 Abu Dhabi Autonomous Racing League; set technical direction across estimation, modeling, and testing to deliver against a fixed deadline under high-speed dynamic constraints.

ENGINEERING EXPERIENCE

Samsung AI Center

Jun 2019 – Aug 2019 | Montreal, QC

Intern — Machine Learning Research Project Support

- **Translated** a novel machine-learning-based resource-allocation algorithm into a testable software prototype, verified in a large-scale network simulation within an automated testing pipeline.

AERO McGill

Sep 2016 – May 2019 | Montreal, QC

Technical Lead

- **Led** a student team of 6 to design and build an autonomous aerial system that won **2nd place** and the **Judges' Award for Professionalism** at the 2019 Unmanned Systems Canada national competition.

OPEN-SOURCE PROJECTS

- `discrete_pid`: published `no_std` Rust PID crate, Simulink-verified, with tests and benchmarks.
- `isaac-drone-racing`: RL drone racing in NVIDIA Isaac Sim (PPO, sim-to-sim transfer, perception-aware reward shaping).

EDUCATION

University of Toronto

2019 – 2025 | Toronto, ON

Ph.D., Aerospace Science & Engineering

Dissertation: “Cooperative Navigation and Control of UAVs in GNSS-Denied Environments.”

McGill University

2015 – 2019 | Montreal, QC

B.Eng., Mechanical Engineering

TECHNICAL SKILLS

State Estimation & Localization: probabilistic robotics, multi-sensor fusion (IMU, LiDAR, GNSS, magnetometer, barometer, motion-capture, range-only), Gaussian estimation, Kalman / Error-State Kalman Filter (ESKF) / nonlinear filtering, smoothing, fault-tolerant estimation & sensor-dropout handling, observability analysis, 3D geometry, geodetic & GNSS-denied localization

Optimization & Math/ML: nonlinear least-squares, factor-graph / batch optimization, numerical & convex optimization, SQP, model-predictive control (Ceres-Solver); linear algebra, calculus, probability theory; machine learning (PyTorch, JAX, CasADi)

Programming: C++ (performance-critical, modern C++17/20), Rust (real-time embedded `no_std`, published crates.io crate), Python, C, Kotlin; memory management, concurrency, real-time computing; Eigen, nalgebra, OpenCV, CMake, pybind11; Linux, Git

Methods & Open Source: first-principles modeling, clean-room & test-driven development, automated evaluation pipelines, hardware-in-the-loop & field validation; merged contributions to Ceres-Solver and ArduPilot

SELECTED PUBLICATIONS

- [1] **H. S. Helson Go**, C. L. Chong, L. Qian, and H. H.-T. Liu. “Observability-Aware Control for Quadrotor Formation Flight with Range-only Measurement.” *AIAA J. Guidance, Control, & Dynamics*, 2026.
- [2] **H. S. Helson Go** and H. H.-T. Liu. “Trajectory Optimization for Cooperatively Localizing Quadrotor UAVs.” *IEEE Int. Conf. on Robotics and Automation (ICRA)*, 2024.
- [3] **H. S. Helson Go**, L. Qian, and H. H.-T. Liu. “Data-Driven and Robust Path-following Control of a Quadrotor Slung Load Transport System.” *AIAA SciTech Forum*, 2023.